

instruction-bounded-inference

On-Chain LLM Inference Under Instruction Budgets: An Instruction-Budget Cost Model, Ternary Floor Evidence, and Session Costs

Authors: Julien Aerni¹, Siméon Fluck², Dustin Becker³

Affiliations: ¹ Meotis Sàrl, Neuchâtel, Switzerland — ORCID 0009-0001-7872-1057. ² Kaizen Corp SA — ORCID 0009-0002-8215-7150. ³ ORIGYN Foundation — ORCID 0000-0002-8505-0597.

Date: May 2026

Categories: cs.DC (primary), cs.LG, cs.CL (cross-list)

ACM CCS: Computing methodologies → Machine learning; Computer systems organization → Distributed architectures.

Keywords: on-chain LLM inference; Internet Computer Protocol (ICP); blockchain AI; decentralized AI; verifiable AI; WebAssembly (WASM) SIMD; deterministic inference; LLM scaling law; instruction budget; quantization; ternary weights; TQ2_0; BitNet; matmul dominance (MUL_MAT); smart-contract inference; stateful inference; agentic AI workloads; small language models (SLM); edge LLM; llama.cpp

TL;DR. On the Internet Computer (ICP), large language model (LLM) inference inside a smart-contract canister is bounded not by hardware throughput but by a deterministic per-call instruction budget of about 40 billion metered units. We calibrate a decode scaling law, $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$; demonstrate a 2.9× software-path speedup on ICP mainnet; produce end-to-end evidence consistent with the ternary $\alpha_{\text{eff}} \approx 1$ instruction-cost floor using a custom TQ2_0 WebAssembly (WASM) SIMD kernel; and model the per-session IO tax that governs stateful, agentic on-chain workloads.

Companion artifact. The local `artifact/` directory (MIT code, CC BY 4.0 data) contains the calibration registry (49+ benchmarks across local dfx, ICP mainnet, and Swiss Subnet), variance-verified anchors, coherence audits, per-operation profiling, multi-call IO fits, source pins (`artifact/notes/source_pinning.md`), and a source-backed execution-boundary note (`artifact/notes/blockchain_ai_execution_boundaries.md`). The companion reproduction repository `instruction-bounded-inference-artifact` contains the end-to-end local rebuild and re-benchmark procedure for the 2.9× fork gap (`REPRODUCE.md`) and its 2026-05-19 result log (`results/rebench_2026-05-19.md`). Every load-bearing numeric claim below cites an explicit CSV or summary-table path; the full claim-to-evidence mapping lives in `CLAIMS-EVIDENCE-MATRIX.md`.

Abstract

LLM inference on Internet Computer Protocol (ICP) canisters runs under a hard per-call budget of $B = 40 \times 10^9$ instruction-cost units, not wall-clock time. Throughput is therefore bounded by executed instructions, not DRAM bandwidth — the *instruction-bounded* regime. The motivating gap: the same Qwen 2.5 0.5B Q8_0 (8-bit uniform integer) model, on the same ICP mainnet canister, yields 10 tok/call on the onicai baseline and 29 tok/call on our fork. The 2.9× gap is software, not hardware.

(R1) Decode cost model. Decode follows $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$, where α_{eff} is the effective cost per FLOP — a deployment-stack property, not a constant. Across 11 modern decoder models from 8 families, $\alpha_{\text{eff}} = 1.527$ (95% bias-corrected bootstrap CI [1.374, 1.65]; leave-one-architecture-out — LOAO — MAPE 7.7%, model-weighted). Legacy GPT-2 code paths form a separate $\alpha \approx 2.0$ regime.

(R2) Matmul dominates. The gap reproduces on mainnet; matmul accounts for 98.7% of decode cost on the profiled Pythia-70M Q8_0 view (98.3% on the Qwen 0.5B Q8_0 SIMD characterization row, §3); per-element unstructured sparsity is provably counter-productive (Theorem 1). Swiss Subnet (SSN) [8] deployments stay within 7% of local calibration.

(R3) Floor evidence. A purpose-built TQ2_0 (2-bit ternary in the GGUF binary format used by llama.cpp [10]) WASM SIMD path on TriLM 560M and TriLM 3.9B both decode at $\alpha_{\text{eff}} \approx 1.00$ — one metered unit per multiply-accumulate, the minimum compatible with the cost model — across a 7× parameter span, with byte-identical outputs on local and 13-replica SSN consensus. We treat this as evidence consistent with the cost-model floor on the measured TQ2_0 path, not a closed proof that all ternary deployments saturate it; the 3.9B point is at integer-token resolution ($N_{\text{MAX}} = 5$).

Three secondary results compose the law into deployment decisions. **(S1)** Uniform Q8_0 stays on the Pareto frontier on SmoLLM2-135M and Qwen 0.5B: no mixed-precision variant strictly dominates it, while Δ PPL vs F16 remains tiny. **(S2)** Multi-call sessions pay a per-call IO tax; a hybrid Mamba+Transformer baseline halves the per-token slope. **(S3)** EmbeddingGemma-300M Q4_0 (4-bit uniform integer GGUF) on SSN runs at $\alpha_{\text{embed}} \approx 0.53$, 126 input tok/call — retrieval fits the regime better than long-form decode.

Plain-Language Summary

Running an AI language model on a blockchain is fundamentally different from running it on a GPU. On the Internet Computer (ICP), a smart contract (a “canister”) gets a fixed, deterministic compute budget per call — roughly 40 billion low-level instructions — and the call fails if it exceeds that budget. This paper measures what that constraint means for large language model (LLM) inference in practice. We find a simple scaling law for how many tokens a model can generate per call; show that rewriting the software path (not changing the hardware) can almost triple output; build a ternary-weight WebAssembly (WASM) kernel that produces end-to-end evidence consistent with the theoretical cost floor; and quantify the extra cost of keeping conversation state across calls, so that on-chain AI agents can be budgeted reliably. The practical recipe: choose the model for quality, choose its parameter count and quantization from the instruction law, optimize the matrix-multiplication (matmul) kernel, and treat saved session state as part of the budget.

1 Introduction: On-Chain LLM Inference as an Instruction-Bounded Problem

1.1 From Blockchain AI to a Measurable Problem

We began from a broader blockchain question: can useful AI inference run *natively on-chain*, inside the consensus boundary, rather than through off-chain workers, oracle submissions, proof systems, or precomputed vectors? In the prior-art review that led to this paper, the relevant systems split into two execution boundaries. Some study verifiability, gas costs, or hybrid on/off-chain execution [14, 19, 20]. Others demonstrate native on-chain inference in specific settings, or propose WASM-based blockchain deployments [9, 21], but without a first-order throughput law for decoder execution inside a consensus-metered runtime.

That boundary distinction became the real problem statement of this paper. We are not asking whether “blockchain AI” exists in some broad orchestration sense; we are asking what happens when the model forward pass itself runs under consensus-enforced compute metering. In this study, ICP is the environment where that question becomes directly measurable: deterministic in-consensus WebAssembly execution [5, 6], an explicit per-message compute budget [7], and enough budget to run sub-billion-parameter models end-to-end. That is why the paper centers on ICP while treating the underlying question as broader than ICP itself.

On native hardware, fixing the model, the quantization, and the machine fixes throughput to within a few percent. On ICP, it does not. Qwen 2.5 0.5B Q8_0 generates **10 tokens per update call** on the onicai fork [9] and **29 tokens per update call** on our fork, despite running in the same mainnet canister framework, on the same GGUF weights, under the same 40-billion-instruction message cap.

Once hardware, weights, and quantization are fixed, the remaining explanation is the software path: kernel choice, SIMD integration, dequantization strategy, and the micro-cost of the executed compute graph. Instruction-metered runtimes [5, 6, 7] are therefore governed by different optimization logic than CPUs and GPUs [1, 2, 25]: the dominant cost is executed instructions, not DRAM bandwidth.

1.2 Four questions

This paper answers four questions in order. Each is addressed in one or two sections, and every load-bearing number resolves to a named CSV row in the companion artifact.

1. **What is the regime?** What distinguishes instruction-metered execution from bandwidth-bound inference, and what cost model follows from it? (§1, §2)
2. **What is the law?** Can decode throughput be predicted from total parameter count with a single stack-dependent coefficient? (§3)
3. **Why should the reader believe it?** Does the law survive production deployment (§4), and can a purpose-built kernel reach its theoretical floor (§5)?
4. **Why does it matter?** How does the per-call law compose into quantization choices and multi-call agentic budgets, and how does the regime extend to non-decode workloads? (§6, §7)

The regime distinction is structural:

	Bandwidth-bound	Instruction-bound
Dominant cost	DRAM bandwidth	Executed instruction count

	Bandwidth-bound	Instruction-bound
Hard budget	Wall-clock (soft)	40 B instructions (hard)
Winning quantization	Fewer bits/weight	Fewer instructions per MAC
<code>f32.mul</code> vs <code>i32.mul</code>	Identical	$2\times$ cost (per ICP cost table)
Main hotspot	Weight loading	Matmul alone (98.3-98.7% across profiled views)

The *qualitative* regime distinction — executed instructions, not DRAM bandwidth, as the binding cost — plausibly extends to other discretely-metered runtimes: blockchain canisters [5, 7], fuelled WASM (Wasmtime `set_fuel`, WasmEdge gas), and possibly zero-knowledge VMs [23, 24]. The *quantitative* results here (β , the `f32/i32` ratio, α_{eff} , the matmul-dominance share) are calibrated to the ICP opcode cost table and are not claimed to transfer numerically; constraint-counting zkVMs in particular have a different cost structure (proving cost can be super-linear in sequence length), where the linear 2P relation need not hold. All measurements in this paper are on the ICP/SSN stack.

1.3 Contributions and related work

We establish three main results (R1–R3) and three secondary results (S1–S3), all traced to the artifact’s 49+ benchmark rows. Prior on-chain ML work [14, 19, 20] includes on-chain inference, gas-cost modelling, and verifiability-oriented system design, but does not provide a first-order scaling law for native instruction-metered decoder throughput on a live ICP/WASM canister stack. Concurrent on-chain inference efforts [9, 21] establish that native deployment is possible, but they describe systems and boundaries rather than a parametric law. The closest prior system is opML [28], which executes an LLM forward pass on-chain inside a fraud-proof VM; it meters execution *lazily* — only when a result is challenged — whereas the ICP setting meters *every* update call eagerly, and our contribution is a throughput cost model rather than a verification protocol. Optimistic-TEE-rollup inference [29] and deterministic-WASM execution [30] target adjacent verifiable-compute boundaries. The novelty claimed here is therefore boundary-specific: native decoder execution under consensus metering, measured as a throughput law, not “blockchain AI” in the broad sense. On-chain embedding offerings typically store precomputed vectors or rely on off-chain workers; our encoder case study (§7) runs the full forward pass inside the consensus boundary and is framed as a systems demonstration, not a claim of priority.

2 The Instruction-Bounded Cost Model for Decoder Inference

ICP canisters execute WebAssembly with deterministic instruction metering [5, 6]. Each opcode has an explicit cost; one update call may consume at most $B = 40 \times 10^9$ ICP cost units [7]. Two asymmetries matter for inference. `f32` arithmetic is priced $2\times$ above `i32` under the ICP opcode cost table, but at the SIMD-kernel level the observed `f32x4` vs `i32x4` accounting is $1.05\times$ because the metering instrumentation block dominates a tight inner loop (`artifact/data/kernel/matmul_bench.csv`). We measure that metering overhead as $\beta = 1.844$ on tight SIMD kernels; β is the metering-overhead multiplier on a tight SIMD inner loop, derived from an empty-loop microbench. The full derivation, and the reconciliation between the $2\times$ opcode-table cost and the $1.05\times$ observed SIMD-kernel ratio, is given in `artifact/notes/beta_derivation.md`.

For decoder-only transformers in the calibrated regime, FLOPs per generated token are well-approximated by 2P [16] (with [1, 2] providing the inference-economics and roofline framing). We decompose per-token cost as

$$C_{\text{token}} \approx \alpha_{\text{eff}} \times 2P,$$

and therefore decode throughput as

$$\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P).$$

We distinguish three levels of the cost coefficient. α_{kernel} is measured on isolated matmul benches with square matrices and random data. α_{eff} is the operational quantity, derived from end-to-end model throughput; it folds in dequantization, dispatch, tiling, metering, and framework overhead ($\alpha_{\text{eff}} > \alpha_{\text{kernel}}$). α_{embed} is the analogous coefficient for encoder forward passes.

α_{eff} is not a constant of nature. It is a property of the deployment stack: family, quantization, compiler, metering, and implementation. On the current stack, α_{eff} resolves into three distinct clusters:

- **Modern decoder families (FP/INT path):** $\alpha_{\text{eff}} \approx 1.53$
- **Legacy GPT-2 code paths:** $\alpha_{\text{eff}} \approx 2.0$ (software-penalty regime)
- **Purpose-built TQ2_0 ternary path:** $\alpha_{\text{eff}} \approx 1.00$ (theoretical floor)

The floor is load-bearing. Setting $\alpha_{\text{eff}} = 1$ corresponds to exactly one metered unit per multiply-accumulate at the 2P count — the minimum compatible with the cost model. No deployment on this stack is observed below it. The claim “the TQ2_0 path reaches the floor” (§5) should therefore be read as measurement evidence that no large remaining kernel-level margin is visible on that measured path at integer-token resolution, not as a proof that every ternary deployment saturates the bound.

3 Empirical Decode Scaling Law for Instruction-Bounded LLM Inference

We calibrate the law on local ICP deployments (dfx 0.31.0 — ICP’s canister CLI — fork build 17806d52), batch size 1, binary search on maximum generated tokens before EXCEEDED_INST_LIMIT. The calibration set is scoped to modern decoder families on the current release line; GPT-2 rows are a legacy sensitivity regime; TQ2_0 rows are discussed in §5 as a distinct regime.

Table 1. Representative calibration rows (artifact/data/models.csv, artifact/data/onchain/local.csv). The first seven rows are the modern-regime calibration examples used to illustrate the main cluster; the final two GPT-2 rows are legacy sensitivity examples shown for contrast and are not part of the modern-only summary statistics reported below. The modern-only fit comprises **eleven models**: these seven rows plus four not shown here for space — Pythia-14M, Mamba-370M, Qwen3-0.6B, and RWKV7-0.4B. Four Transformer anchor points (†) are variance-verified with 3 repetitions each (CV < 1.05%); a fifth variance-verified point, Mamba-370M, is **retained in the modern fit** — it carries the largest per-architecture LOAO error (18.3%, reported in §3) — not excluded. All other rows are single-run binary-search estimates.

Model	Arch	Params	tok/call	α_{eff}	Notes
Pythia-70M†	gpt_neox	70 M	208	1.37	Q4_0 SIMD
SmolLM2-135M†	llama	135 M	97	1.53	Q4_0 SIMD
Gemma3-270M-IT†	gemma3	270 M	56	1.32	Q4_0 SIMD
OpenELM-270M	openelm	270 M	45	1.65	Q4_0 SIMD
SmolLM2-360M	llama	360 M	36	1.54	Q4_0 SIMD
Qwen 2.5 0.5B†	qwen2	494 M	31	1.31	Q4_0 SIMD
H2O-Danube3-500M	llama	500 M	27	1.48	Q4_0 SIMD
DistilGPT2	gpt2	82 M	119	2.05 $\triangle$	legacy regime
GPT-2-124M	gpt2	124 M	84	1.92 $\triangle$	legacy regime

Table 1b. Network and deployment rows measured on ICP mainnet or Swiss Subnet (SSN). These rows are included in the artifact, but they are not all used in the modern-only scaling-law fit above: some are same-build network confirmations, some are build-specific deployment observations, some are systems demonstrations, and some are negative/operational evidence.

Model / setup	Env	Quant / mode	tok/call or outcome	Role in paper
Qwen 2.5 0.5B, our fork	ICP main net	Q8_0	29	Production 2.9× software-gap result
Qwen 2.5 0.5B, onicai baseline	ICP main net	Q8_0	10	Baseline for same canister / weights
SmolLM2-135M	SSN	Q4_0	97	Same-band network validation
Gemma3-270M-IT	SSN	Q4_0	52	Same-band network validation (−7% vs local)
Qwen 2.5 0.5B	SSN	Q4_0	30	Same-band network validation (−3% vs local)
Qwen3.5 0.8B	SSN	Q4_0	13	Highest-quality SSN decoder row; crashes ICP mainnet
Qwen3.5 0.8B	SSN	TQ2_0	18	Throughput valid, quality invalid; negative result
Qwen3.5 0.8B-21L	SSN	Q4_0	18	Pruning throughput valid, quality invalid
Qwen3.5 0.8B-12L	SSN	Q4_0	20	Pruning throughput valid, quality invalid
Qwen3.5 0.8B-12L	SSN	TQ2_0	25	Retracted throughput gain: quality invalid
Qwen3.5 0.8B	SSN	IQ2_XXS	5	Aggressive quantization; not used as evidence
Falcon-H1-Tiny-90M	SSN	Q8_0	199	Build-specific hybrid Mamba+Transformer observation; used for §6.2 IO analysis, not same-build calibration
TriLM 560M	SSN	TQ2_0	34	Ternary floor validation, 3 reps, byte-identical
TriLM 3.9B	SSN	TQ2_0	5	Ternary floor validation, 3 reps, byte-identical
SmolLM2-135M	SSN	BASE-Q8_0	95	Paper 1.5 C2 Pareto-winner cross-env validation
Qwen 2.5 0.5B	SSN	BASE-Q8_0	28	Paper 1.5 C2 Pareto-winner cross-env validation
Qwen 2.5 0.5B	SSN	BASE-Q8_0, 10-turn	100 total tokens	Paper 1.5 C3 multi-call session demo
EmbeddingGemma-30 0M	ICP / SSN	Q4_0 encoder	126 input tok/call	Encoder systems demonstration

The full network tables are artifact/data/onchain/icp_mainnet.csv and artifact/data/onchain/ssn_mainnet.csv; Paper 1.5 deployment rows are also summarized in artifact/data/paper_1_5/

ternary_measurements.csv, mixed_precision_measurements.csv, and multicall_characterization.csv.

Table 1c. Additional Qwen 2.5 0.5B canister characterization retained from the long Paper 1 draft. These rows explain the software gap and prompt-length limits; they are not additional same-status calibration anchors.

Measurement	Value	Interpretation
Q8_0 SIMD, local canister	30 tok/call	Same optimized path as the 29 tok/call mainnet row, within build/network drift
Q8_0 scalar, local canister	6 tok/call	Scalar path is practically unusable at 0.5B scale
F16, local canister	~3 tok/call	No F16/F32 SIMD path; quantized SIMD is required
Decode profiling, Q8_0 SIMD	12.327B instructions / 10 tokens	$\alpha_{\text{eff}} = 1.23$ in the optimized path
MUL_MAT share, same profile	98.3%	The Qwen profile independently confirms matmul dominance
Estimated scalar α_{eff}	~6.75	SIMD compresses effective cost by about 5.5× for Qwen
Prompt ~5 tokens	0 prefill calls; ~28–30 gen tok/call	Short guardrail prompts can stay single-call
Prompt ~50 tokens	2 prefill calls; ~20 gen tok/call	Chat-style prompts pay visible multi-call prefill cost
Prompt ~120 tokens	4 prefill calls; ~10–17 gen tok/call	Long prompts are latency-bound before generation

The underlying registry rows are in artifact/results/raw/core_measurements_v2.csv; the network rows are in artifact/data/onchain/. The long-form measurement inventory is preserved in artifact/notes/paper1_long_measurement_inventory.md for reviewers who want the distinction between fit rows, deployment rows, negative results, and canister-characterization rows.

Fit statistics on the modern-only regime (artifact/results/current/scaling_law/, via scripts/analyze_scaling_law.py):

- Median $\alpha_{\text{eff}} = 1.527$, 95% bias-corrected bootstrap CI [1.374, 1.65] (BCa, 9 999 resamples)
- Strict modern-only **LOAO MAPE** = 7.7% (model-weighted; family-equal weighting gives 8.9%), max per-architecture error 18.3%
- Log-log slope **-0.980**, 95% OLS CI [-1.047, -0.913], R^2 0.9919 — consistent with the -1 expected from $\text{tok/call} \propto 1/(\alpha_{\text{eff}} \cdot 2P)$

Legacy GPT-2 rows land at $\alpha \approx 2.0$ — systematically above the modern cluster, and only on ICP. On native Apple M4 Max the same models show no regime split (artifact/data/native/m4_max_baseline.csv); the penalty lives in the llama.cpp GPT-2 code path, not in the architecture itself.

Architecture details (depth, hidden size, vocabulary) do not independently predict throughput once total parameter count is fixed. Danube3-500M and Qwen 2.5 0.5B occupy very different layer/hidden geometries but share a throughput band because they share a parameter band. Non-Transformer coverage is preliminary: RWKV-7 [12] lands consistent with the cluster; Mamba-370M [13] drifts between builds ($\alpha = 1.54 \rightarrow 1.80$, -14% tok/call) and we treat its result as suggestive, not settled.

Sensitivity to the parameter convention. 2P counts only trainable linear weights in our fits. GGUF-audited parameter counts occasionally exceed nominal counts by 15–25 % (e.g. Qwen 2.5 0.5B: 494 M nominal vs 630 M audited; cf. CLAI MS-EVIDENCE-MATRIX.md discrepancies log). Recalibrating on audited counts compresses α_{eff} by a similar margin without changing the modern-regime cluster or the $\alpha \approx 1$ floor.

Absolute tok/call numbers are build-sensitive within $\pm 15\%$ between March and April 2026 builds. The scaling-law *structure* — slope, model ordering, α_{eff} median — is stable across those builds; calibration points must be tied to a concrete WASM hash.

4 Production Validation on ICP Mainnet and Matmul (MUL_MAT) Dominance

4.1 The 2.9× fork gap on mainnet

We measured the same Qwen 2.5 0.5B Instruct Q8_0 GGUF (SHA-256 ca59ca7...) on ICP mainnet canister zmm32-7yaa-aaaad-qlsq-qai under the same prompt and the same 40 B per-call instruction cap:

Fork	WASM hash	max gen tok/call	Result
Ours	ef8f9d78...	29	OK
onikai [9]	6c77a958...	10	OK

The 2.9× gap is a production result, not a local artefact (artifact/data/onchain/icp_mainnet.csv). Identical weights and identical boundary produce a 2.9× throughput difference only because the software path differs.

Independent re-validation (2026-05-19). The functional claim was re-validated on a local dfx replica using freshly rebuilt WASMs from current public sources: Comoto fork at `Simlowker/gian@8cda13b` (rebuild SHA-256 `da112d99...`) and onicai baseline at `onicai/llama_cpp_canister@v0.10.1` (rebuild SHA-256 `b6ccbff0...`). Using the identical Qwen 2.5 0.5B Q8_0 GGUF (SHA-256 `ca59ca7f...`) and identical prompt and protocol, the rebuild yields **Comoto = 29 tok/call, onicai = 10 tok/call, ratio = 2.9×** — exactly matching the original 2026-04-09 mainnet measurement. The original WASM `ef8f9d78...` is not byte-reproducible: it was built from working-tree state that was never committed to git (a `.gitignore` pattern shadowed 114 per-model `.cpp` files in `src/models/`; commit `668b48e` later imported them as a corrective fix). Byte-exact rebuild is therefore impossible, but functional behavior — the claim itself — is fully reproducible from current public source by any reviewer with dfx, ~50 GB disk, and ~30 min CPU; no ICP cycles required. The end-to-end procedure is in the companion `instruction-bounded-inference-artifact` repository, file `REPRODUCE.md`; the reproduced 29-vs-10 result is logged in `results/rebench_2026-05-19.md` in that repository.

4.2 Matmul dominates at 98.7%

Per-operation profiling at ggml-operation boundaries on Pythia-70M Q8_0 decode (`artifact/data/profiling/per_operation.csv`) attributes:

Operation class	Share of total cost
MUL_MAT	98.7%
All non-matmul work combined	1.3%

This sharpens the optimization picture. Integer normalization, alternate attention formulations, and other non-matmul tweaks target $\leq 1.3\%$ of cost on the profiled Pythia-70M Q8_0 view ($\leq 1.7\%$ on the Qwen 0.5B Q8_0 SIMD profile reported in §3); kernel-table opcode differences (`f32` vs `i32`) are drowned by metering overhead $\beta = 1.844$ (`artifact/notes/beta_derivation.md`). The only lever that moves end-to-end throughput is effective matmul cost. Matmul-free language modeling [18] is an orthogonal architectural response to the same dominance pattern; for a decoder running in our metering regime, eliminating matmul opcodes outright would change the cost decomposition itself rather than reduce the coefficient.

4.3 Why unstructured sparsity fails (Theorem 1)

Theorem 1. *For any sparsity rate $s \in [0, 1]$, per-element zero-checking strictly increases total instruction cost relative to a sparsity-oblivious matmul on the integer-matmul ICP WebAssembly paths measured here (`TQ2_0`, `Q4_0`, `Q8_0`).*

Proof sketch. The `i32.load` of `w_i` is required by both the baseline MAC and the sparsity check, so it is shared and not counted as overhead; the check inserts `i32.eqz (1) + br_if (2) = 3` additional ICP units per weight. If `w_i = 0`, the check skips a MAC whose cost is 1 unit on the `TQ2_0` ternary path (heavier on `Q8_0/Q4_0` where the saving stays ≤ 2). Net per zero on `TQ2_0`: $+3 - 1 = +2$; net per non-zero: $+3$. Total additional cost on the ternary path: $N \cdot (3 - s) > 0$ for all $s \leq 1$. The same strict positivity holds a fortiori on heavier-MAC integer paths where the per-zero saving never matches the $+3$ overhead. The SIMD variant (16-lane zero-check) and the branchless `select` variant both remain strictly positive under analogous counting (`artifact/data/kernel/matmul_bench.csv`). $\square$

Scope. The theorem covers per-element branching, SIMD all-zero checking, and branchless `select` on the integer-matmul paths in this study. It does not apply to *structured* sparsity (N:M patterns with block-level skipping), compressed formats (CSR/CSC), or bitmap indexing that amortizes the check cost. F32-path sparsity is also out of scope: `f32` opcode pricing differs and the per-MAC saving structure changes.

4.4 Cross-subnet consistency

SSN Q4_0 deployments (`artifact/data/onchain/ssn_mainnet.csv`) stay inside the calibration band: SmolLM2-135M matches local at 97 tok/call, Gemma3-270M within 7%, Qwen 2.5 0.5B within 3%. These deltas are consistent with ordinary build and deployment variance, not a different governing law.

5 Reaching the Ternary $\alpha_{\text{eff}} \approx 1$ Instruction-Cost Floor (`TQ2_0`)

5.1 TriLM ternary at 560 M and 3.9 B both hit $\alpha_{\text{eff}} \approx 1.00$

We integrated a custom `TQ2_0` (2-bit ternary, weights in $\{-1, 0, +1\}$) WASM SIMD path — absent from upstream `llama.cpp` [10] — end-to-end into the inference framework, and measured two purpose-trained ternary models from the SpectraSuite TriLM family [15] on the same canister build (`artifact/data/paper_1_5/ternary_measurements.csv`, `artifact/results/paper_1_5/PHASE-1-SUMMARY.md`). The broader ternary-LLM line — BitNet-style 1-bit and 1.58-bit training [3, 22], ternary sparsification [17], edge-inference kernels [4], and CPU

SIMD reorganizations for ternary inference [27] — motivates the integer-matmul boundary we exploit here; we differ in targeting a consensus-metered WASM canister rather than native hardware:

Metric	TriLM 560M	TriLM 3.9B
Params (GGUF-audited)	569 M	3 992 M
GGUF size (TQ2_0)	195 MB	1 113 MB
Architecture	24 L × 1280 H	30 L × 3072 H
N_MAX local dfx (3 reps)	34 (CV 0%)	5 (CV 0%)
N_MAX SSN mainnet (3 reps)	34 (CV 0%)	5 (CV 0%)
Implied α_{eff}	1.03	1.00
Paper 1 modern-arch baseline ($\alpha=1.54$)	22.8	3.3
Speedup vs baseline	1.49×	1.50×
% of theoretical floor ($\alpha = 1$)	97 %	≈100 % (N=5, ±20 % res.)
Coherence	10/10 fluency	factual ✓ at N=5

Two points $7\times$ apart in parameter count both land at $\alpha_{\text{eff}} \approx 1$, the cost-model floor for this path. We read this as evidence consistent with the floor on the measured TQ2_0 path; the 3.9B point sits at integer-token resolution $N_{\text{MAX}} = 5$ ($\approx 20\%$ per-step granularity in α), so we do not claim closed saturation. The 1.49–1.50 $\times$ speedup over the modern-arch baseline is constant across the $7\times$ span, weakening the alternative explanation that the 560 M result was inflated by sub- $\alpha = 1.54$ inefficiency at small scale.

5.2 Byte-exact cross-environment determinism

A stronger consistency check than throughput matching is bit-for-bit output agreement. We re-ran the 10-prompt TriLM 560M coherence audit on the 13-node SSN canister and diffed completions byte-by-byte against the local dfx run (`artifact/results/paper_1_5/raw/trilm-560m-ssn-vs-mainnet-diff.txt`): **empty diff — all 10 completions byte-identical**. For TriLM 3.9B every token produced at $N = 1..5$ on SSN matches the local output verbatim. This rules out non-deterministic kernel hazards (uninitialized vector lanes, FP reorder, replica-specific code paths). The TQ2_0 WASM SIMD kernel is fully deterministic at both scales under live 13-replica consensus.

5.3 Throughput density

On non-purpose-trained ternary weights, the same TQ2_0 path reaches **201 tok/call on Pythia-70M**, giving **5.03 tok/MB** — the highest throughput-per-megabyte observed in the study (`artifact/results/current/extended_analysis/`). Q4_0 SIMD remains the best absolute-throughput operating point for non-ternary models in the calibration set. A ternary-native DOT $\times 4$ microkernel projects a 2.45 $\times$ bench-level speedup over F32 SIMD (artifact only; no model-level run yet).

6 Pareto Quantization (Mixed-Precision vs Uniform Q8_0) and the Multi-Call Session IO Tax

6.1 Q8_0 stays on the Pareto frontier

The per-call law does not by itself say which quantization to pick. We computed per-layer ΔPPL sensitivity on WikiText-2, generated six mixed-precision variants per model (embedding $\rightarrow$ Q8, attention $\rightarrow$ Q8, first K layers $\rightarrow$ Q8, top- ΔPPL layers $\rightarrow$ Q8) on a Q4_K_M base, and compared against uniform Q4_0, Q5_K_M, Q8_0 on SmolLM2-135M and Qwen 2.5 0.5B. Throughput rows in `artifact/data/paper_1_5/mixed_precision_measurements.csv`; perplexity rows in `c2-ppl-*.csv`; joined summary in `artifact/results/paper_1_5/tables/c2-mixed-precision-summary.md`.

Table 2. Pareto winners on both models. Qwen 0.5B Q8_0 size is reported as the Phase 2 local build (506.5 MB, matching the anchored throughput row); the Phase 3 SSN rebuild is 531 MB with byte-exact weights, differing only in metadata.

Metric	SmolLM2-135M	Qwen 2.5 0.5B
Preferred frontier point	BASE-Q8_0	BASE-Q8_0
size (MB)	144.8	506.5
tok/call	97	30
PPL WikiText-2	13.603	11.680
PPL WT2 (F16 baseline)	13.574	11.674
ΔPPL vs F16	+0.029	+0.007
$\alpha_{\text{eff}} = B / (N \cdot 2P)$	1.527	1.333

Uniform Q8_0 is the highest-throughput, lowest-PPL point on both tested frontiers, but not the smallest. The precise statement is: no mixed variant strictly dominates it. Q5_K_M is the sole strictly-dominated variant. The ΔPPL -guided V4

variant (top-K fragility layers at Q8, rest at Q4_K_M) is the best mixed variant and remains relevant under tight size budgets.

Two caveats. First, neither hidden dim (576, 896) is divisible by 256, so llama-quantize emits many q4_K/q5_K → q5_0/q8_0 fallbacks; the mixed-precision landscape likely differs on hidden dims in {512, 1024, 2048}. Second, Qwen’s $\alpha_{\text{eff}} = 1.33$ is lower than SmolLM’s 1.53 despite larger hidden dim. Plausible contributors: GQA (grouped-query attention; fewer K/V FLOPs), attention biases not counted in 2P, and SmolLM2’s tied embedding making 2P undercount its LM head. Both values remain inside the Paper 1 modern-regime CI.

Validated on SSN. Pareto-winner BASE-Q8_0 on SSN mainnet: SmolLM2-135M N_MAX = 95 (local = 97, −2%, byte-exact 3/3 reps); Qwen 2.5 0.5B N_MAX = 28 (local = 30, −6.7%, byte-exact 3/3 reps). Source: artifact/results/paper_1_5/tables/c2-c3-ssn-validation-summary.md.

6.2 Multi-call stateful session budget

A single 40 B call is often not enough for agentic workloads. Using --prompt-cache-all, a session of N calls pays two extra per-call IO costs the single-call formula ignores: save_instr (serialize the growing KV cache at end-of-call) and load_instr (deserialize at start of calls 2..N), both in raw instructions. These reduce the effective per-call decode budget, giving an **extended session formula**:

$$\text{tok/call}_{\text{effective}} \approx (\text{B} - \text{IO}_{\text{instr}}(\text{n}_{\text{kv}}, \text{cache_type})) / (\alpha_{\text{eff}} \times 2\text{P})$$

$$\text{tok/session} \approx \text{N} \times \text{tok/call}_{\text{effective}}$$

with $\text{IO}_{\text{instr}}(\text{n}, \text{c}) = \text{save_instr}(\text{n}, \text{c}) + \text{load_instr}(\text{n}, \text{c}) \approx (\text{a}_s + \text{b}_s \cdot \text{n}) + (\text{a}_l + \text{b}_l \cdot \text{n})$ fitted from instrumented canister runs (ggml commit bb238b9 adds a C3-IO-END log line exposing save_instr and load_instr per call). Units: B and IO_instr in ICP instruction-cost units; $\alpha_{\text{eff}} \times 2\text{P}$ in instructions per token.

Table 3. Fitted IO parameters (artifact/data/paper_1_5/multicall_characterization.csv, artifact/results/paper_1_5/tables/c3-fit-parameters.json).

Model / cache (f16)	a_s (M)	b_s (M/tok)	R ²	load mean (M)	load CV
Qwen 2.5 0.5B Q4_0 (Transformer)	26.7	0.951	0.998	425.6	0.02%
Falcon-H1-Tiny-90M Q8_0 (hybrid Mamba+T)	276.3	0.519	0.972	108.3	3%

Observations. The hybrid Mamba+Transformer baseline (Falcon-H1-Tiny, in the broader family of hybrid-head SSM/attention small models [26]) halves the per-token save slope (0.52 M vs 0.95 M) and cuts the fixed load cost 4× (108 M vs 426 M), but does not eliminate per-token growth — attention blocks still accumulate KV. The SSM advantage is partial, not the full “bounded state” headline. --cache-type-k {f16, q8_0, q4_0} is accepted at load_model but is a no-op on the current WASM (KV allocated before the flag reaches the cache-swap path); a rebuild exposing it at runtime is the obvious next lever. Re-attach latency is dominated by load_model itself (~2.3 s wall-clock), not by state IO (artifact/results/paper_1_5/raw/c3-reattach-qwen05.log). A 10-turn chat demo on SSN (Qwen 2.5 0.5B Q8_0) completed 10/10 turns, 100 cumulative tokens, 41.12 s wall-clock, with narrative coherence preserved under 13-node consensus (c3-ssn-chat-demo-transcript.md).

For short-turn agents (cache ≪ 100 tok) the constant term dominates and Falcon-H1 wins the cold path by ~15%; for medium sessions (cache ~100 tok) by ~20%; for long chat (cache ≫ 500 tok) the slope difference becomes load-bearing. The extended session formula is the operative budget for any on-canister agentic workload.

6.3 Operational ceilings

Two ceilings bound the deployable envelope. First, on the tested SSN wasm32 build we observed a **practical wasm-heap ceiling near 900 MB**; in the Qwen 0.5B Q8_0 load path, reducing to -c 512 -b 512 (context and batch) brought the allocation under that envelope and loaded cleanly. The background for non-ICP readers: a wasm32 canister heap is bounded by a 32-bit WebAssembly address space (theoretical 4 GiB), while stable memory is a separate persistent store and current ICP documentation also distinguishes wasm64 limits. The observed ceiling therefore reflects current heap-side allocations — weights staging, KV cache, compute buffers, framework state, metering instrumentation, and scratch space — rather than a single pool shared with stable memory. Treat 900 MB as a build-specific operational observation, not a universal threshold. Second, an IC0524 **page-access trap** triggers on some ≥ 1 GB GGUFs during load_model, but not on raw size alone: TriLM 3.9B TQ2_0 at 1 113 MB loads cleanly, while Qwen 2.5 1.5B Q6_K at 1 464 MB traps IC0524 (artifact/data/paper_1_5/ic0524_characterization.csv). The failure is format-specific — ternary TQ2_0 is unusually friendly to the ICP page-access pattern. A parallel IC0502 “unreachable” trap at different sizes (Qwen 1.5B Q4_0 at 892 MB) is driven by Wasm-runtime limits rather than stable-memory limits; the two failure modes should not be conflated.

7 Secondary Results (On-Chain Encoder Case Study) and Limitations

7.1 On-chain encoder case study

EmbeddingGemma-300M Q4_0 [11] runs end-to-end inside an SSN canister (u2pva-3iaaa-aaaax-qaa7a-cai, 212 MB on disk) at about **126 input tokens per call**, corresponding to $\alpha_{\text{embed}} \approx 0.53$ (artifact/data/onchain/ssn_mainnet.csv embedding rows). That is materially cheaper per input token than autoregressive decode, suggesting retrieval-style workloads fit the instruction-bounded regime better than long-form chat generation. The deployment is a systems demonstration (N = 1 model, small cross-lingual sanity checks). It is not a claim of priority on the broader “on-chain embedding” term: ecosystem projects under different system boundaries — stored precomputed vectors, off-chain workers, oracle submissions — have existed for some time.

7.2 Minor systems contribution

A WASM SIMD vectorized unpack for the Q6_K q1/qh path (reviewer-flagged scalar path in upstream PR #11453) gives +2.3% matmul-bench speedup and 3.2× variance reduction (CV 8.0% → 2.5%, bit-exact vs the scalar reference). End-to-end on canister the gain stays below integer-token granularity and is kept in the artifact for future Q6_K workloads that might land above the threshold.

7.3 Limitations

1. **Calibration coverage.** The modern-regime law ($\alpha_{\text{eff}} \approx 1.53$) is anchored on the 70 M–800 M range; the ternary floor ($\alpha_{\text{eff}} \approx 1.00$) is validated at 560 M and 3.9 B but only on the TriLM family and only for TQ2_0. Neither result is yet a universal law across architecture, quantization, and scale.
2. **Variance coverage.** 4 of 17 modern-Transformer calibration points are variance-verified (3 reps, CV < 1.05%) — Pythia-70M Q4_0, SmolLM2-135M Q4_0, Gemma3-270M-IT Q4_0, Qwen 2.5 0.5B Q4_0; a fifth variance-verified point, Mamba-370M, is the non-Transformer baseline included in the modern-only LOAO fit, where it is the single largest per-architecture error (18.3%). The remaining single-run binary-search estimates still contribute to the LOAO summary and the reported modern-only BCa interval in §3, so the interval should be read as conditional on a mixed repeated/single-run evidence base rather than as a repetition-only uncertainty estimate.
3. **Non-Transformer evidence is preliminary.** RWKV-7 is consistent; Mamba is build-sensitive. Cross-architecture generalization is not settled.
4. **Build sensitivity.** Absolute tok/call drifts within $\pm 15\%$ between builds; calibration points must be tied to a WASM hash.
5. **Prefill.** Long-prompt usability remains poor under a per-call budget: a 512-token SmolLM2-135M prompt requires ~15 sequential calls (~15–30 s latency) before the first generated token. Decode may obey a simple law while user-visible latency still needs multi-call orchestration.
6. **Throughput is not quality.** Higher tok/call does not imply a better model; it implies a cheaper one under the budget. We retain Qwen 3.5 ternary and layer-pruned variants in the artifact as explicit negative results: throughput-valid but quality-invalid (artifact/data/quality_audit.csv, flagged role=negative_result).
7. **Embedding evaluation.** The encoder case study is N = 1 with small sanity checks; MTEB-grade on-chain evaluation is a template (scripts/run_mteb_fra.py), not a completed benchmark.

8 Conclusion

Instruction-metered execution creates a distinct inference regime. On ICP, decode throughput is not best understood through bandwidth language; it is a budgeted compute problem with a hard per-call cap and a stack-dependent effective cost coefficient.

Within that regime we establish three facts on the current release line. A simple first-order law $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$ holds for modern decoder families, with a separate penalty regime for legacy GPT-2 code paths. The production explanation is software: the fork gap reproduces on mainnet, matmul is the only first-order optimization lever, and per-element sparsity is formally excluded. Evidence consistent with the cost-model floor is observed on a purpose-built kernel and purpose-trained ternary weights at both 560 M and 3.9 B, under live 13-replica consensus, with byte-identical outputs across environments — a systems result, not a universal theorem of saturation.

Two derived results matter for deployment. Uniform Q8_0 remains non-dominated against the mixed-precision variants tested, and multi-call stateful inference pays a predictable two-term IO tax per call. The practical consequence is narrow: choose architecture for quality; choose parameter count and quantization from the law; treat matmul-path engineering as the primary throughput lever; and budget IO overhead once sessions exceed a single call. The 40 B instruction budget is not the operative deployment ceiling at current scales — wasm heap and page-access limits are. Extensions to larger scales and to ternary-native DOT ×4 at model level are straightforward follow-ups, already scoped in the artifact scripts.

Data and Code Availability

All code, data, rebuild scripts, and raw logs behind every load-bearing claim are released as a companion artifact: code under the MIT License, data under CC BY 4.0. The claim-to-evidence mapping is `CLAIMS-EVIDENCE-MATRIX.md`; the readiness check artifact/scripts/check_paper_readiness.py validates registry/table/claim consistency.

- **Companion artifact** (calibration registry, per-operation profiling, Paper 1.5 CSVs, scripts, exports): the artifact / directory of this package.
- **Reproduction repository**: `github.com/Simlowker/instruction-bounded-inference-artifact` — the end-to-end local rebuild of the $2.9\times$ fork gap (`REPRODUCE.md`) and its result log (`results/rebench_2026-05-19.md`). Made public at deposit.
- **Archived snapshot + DOI**: Zenodo 10.5281/zenodo.20607598 (reserved at deposit; the GitHub release is archived via the Zenodo–GitHub integration).
- **Live deployments**: ICP mainnet decode canister `zmm32-7yaaa-aaaad-qlqsq-cai`; SSN encoder canister `u2pva-3iaaa-aaaax-qaa7a-cai`.

Table DA1. Key artifacts → location → SHA-256 → claim supported.

Artifact	Location	SHA-256 (abridged)	Supports
Qwen 2.5 0.5B Q8_0 GGUF	instruction-bounded-inference-artifact/REPRODUCE.md	ca59ca7f...6844e	§4.1 mainnet 29-vs-10
Comoto fork WASM (rebuilt, Simlowker/gian@8cda13b)	results/rebench_2026-05-19.md	da112d99...a49c	§4.1 (29 tok/call)
onikai WASM (rebuilt, onikai/llama_cpp_canister@v0.10.1)	results/rebench_2026-05-19.md	b6ccbff0...9660	§4.1 (10 tok/call)
Calibration registry (49 rows)	artifact/data/models.csv, artifact/data/onchain/	—	§3 scaling law
Scaling-law outputs (BCa, LOAO)	artifact/results/current/scaling_law/	—	§3 ($\alpha_{\text{eff}} = 1.527$)
Ternary floor (TriLM 560M/3.9B)	artifact/data/paper_1_5/ternary_measurements.csv	—	§5 ($\alpha_{\text{eff}} \approx 1$)
Mixed-precision sweep	artifact/data/paper_1_5/mixed_precision_measurements.csv	—	§6.1 Pareto
Multi-call IO fits	artifact/data/paper_1_5/multicall_characterization.csv	—	§6.2 IO tax
Operational ceilings (IC0524)	artifact/data/paper_1_5/ic0524_characterization.csv	—	§6.3

Full SHA-256 values, rebuild commits, baseline tags, and the non-reproducible original-WASM caveat are in `artifact/notes/source_pinning.md`.

References

- [1] T. Wu et al., “Inference Scaling Laws: An Empirical Analysis of Compute-Optimal Inference for LLM Problem-Solving,” arXiv:2408.00724.
- [2] Z. Yuan et al., “LLM Inference Unveiled: Survey and Roofline Model Insights,” arXiv:2402.16363.
- [3] S. Ma et al., “The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits,” arXiv:2402.17764.
- [4] J. Wang et al., “Bitnet.cpp: Efficient Edge Inference for Ternary LLMs,” arXiv:2502.11880.
- [5] DFINITY Foundation, “IC Interface Specification,” docs.internetcomputer.org/references/ic-interface-spec. Accessed April 2026.
- [6] DFINITY, “IC WASM Instrumentation,” github.com/dfinity/ic/blob/master/rs/embedders/src/wasm_utils/instrumentation.rs (source-provenance note in artifact/notes/source_pinning.md). Accessed April 2026.
- [7] DFINITY Foundation, “IC Per-Message Instruction Limits,” documented in the IC Interface Specification (docs.internetcomputer.org/references/ic-interface-spec, “Nomenclature: trap on instruction limit”). Accessed April 2026.
- [8] DFINITY Foundation, “ICP Swiss Subnet,” NNS Proposal 136408, dashboard.internetcomputer.org/proposal/136408; Swiss Subnet introduction forum thread, forum.dfinity.org/t/introducing-the-swiss-

subnet-a-secure-and-compliant-blockchain-infrastructure-leveraging-the-internet-computer-protocol/41135. Accessed April 2026.

[9] onicai, “llama_cpp_canister,” github.com/onicai/llama_cpp_canister, baseline release tag v0.9.0 (github.com/onicai/llama_cpp_canister/releases/tag/v0.9.0); re-validated against v0.10.1 on 2026-05-19 (post-v0.9.0 commits are housekeeping only, no inference-path change).

[10] ggml-org, “llama.cpp: LLM inference in C/C++,” github.com/ggml-org/llama.cpp (source-provenance note in [artifact/notes/source_pinning.md](https://github.com/ggml-org/llama.cpp/blob/main/artifact/notes/source_pinning.md)). Accessed April 2026.

[11] Google, “EmbeddingGemma: Powerful and Lightweight Text Representations,” arXiv:2509.20354.

[12] B. Peng et al., “RWKV-7 ‘Goose’ with Expressive Dynamic State Evolution,” arXiv:2503.14456.

[13] A. Gu, T. Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” arXiv:2312.00752.

[14] A. Alhaidari et al., “On-Chain Decentralized Learning and Cost-Effective Inference for DeFi Attack Mitigation,” arXiv:2510.16024.

[15] A. Kaushal et al., “Spectra: Surprising Effectiveness of Pretraining Ternary Language Models at Scale,” arXiv:2407.12327 (TriLM suite source).

[16] N. Sardana, J. Portes, S. Doubov, J. Frankle, “Beyond Chinchilla-Optimal: Accounting for Inference in Language Model Scaling Laws,” arXiv:2401.00448.

[17] H. Huang et al., “Sherry: Hardware-Efficient 1.25-Bit Ternary Quantization via Fine-grained Sparsification,” arXiv:2601.07892.

[18] R. Zhu et al., “Scalable MatMul-free Language Modeling,” arXiv:2406.02528.

[19] Z. Li, S. Vott, B. Krishnamachari, “ML2SC: Deploying Machine Learning Models as Smart Contracts on the Blockchain,” arXiv:2404.16967.

[20] R. Karanjai, L. Xu, W. Shi, “Weaving the Cosmos: WASM-Powered Interchain Communication for AI Enabled Smart Contracts,” arXiv:2502.17604.

[21] DFINITY Foundation forum, “Walsh: Hypercomplex LLM Inference on the IC (1.58-bit Quantized),” forum.dfinity.org/t/walsh-hypercomplex-llm-inference-on-the-ic-1-58-bit-quantized/61676. Accessed April 2026.

[22] Microsoft Research, “BitNet b1.58 2B4T Technical Report,” arXiv:2504.12285.

[23] T. Gassmann, S. Chaliasos, T. Sotiropoulos, and Z. Su, “Evaluating Compiler Optimization Impacts on zkVM Performance,” arXiv:2508.17518.

[24] T. South et al., “Verifiable Evaluations of Machine Learning Models via zkSNARKs,” arXiv:2402.02675.

[25] E. Erdil, “Inference Economics of Language Models,” arXiv:2506.04645.

[26] X. Dong et al., “Hymba: A Hybrid-head Architecture for Small Language Models,” arXiv:2411.13676.

[27] H. Oh et al., “T-SAR: A Full-Stack Co-design for CPU-Only Ternary LLM Inference via In-Place SIMD ALU Reorganization,” arXiv:2511.13676.

[28] KD Conway, C. So, X. Yu, K. Wong, “opML: Optimistic Machine Learning on Blockchain,” arXiv:2401.17555.

[29] A. Chan, A. Ding, F. Chen, A. Wu, “Optimistic TEE-Rollups: A Hybrid Architecture for Scalable and Verifiable Generative AI Inference on Blockchain,” arXiv:2512.20176.

[30] W. Zhou, X. Xu, C. Wei, Y. Yan, “DTVM: Revolutionizing Smart Contract Execution with Determinism,” arXiv:2504.16552.